William Shen

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Technical Skills

Languages: Java, Python, C/C++/C#, SQL, JS/TS, HTML/CSS, R, Powershell
Frameworks: React, Node.js, Flask, Next.js, Firebase, Tailwind
Developer Tools: Git, Docker, Google Cloud Platform, Jira

Education

University of Waterloo — Waterloo, ON
Bachelor of Mathematics in Computational Mathematics — *Sep 2023 – Present*
Minor in Combinatorics & Optimization

Experience

Software Developer — May 2026 – Present
Geotab — *Waterloo, ON*

- Led migration of the MyGeotab API Reference from legacy **XML** file to the new **JSON** source, rewriting the parser in **Typescript** while validating full coverage across **500+** objects and methods
- Integrated with Vodafone, adding "Report as Stolen" button in **React** frontend and annual fleet enrollment renewal automation using **C#** against Vodafone APIs to save **$135K+/yr** in subscription costs and ensure accuracy
- Integrated with McLeod TMS, shipping "Connect" **UI** and functionality, and building a production **Flink** pipeline filtering telemetry across 3 **Kafka** topics by company GUID for real-time sync
- Developed a **28**-chart **Superset** tab for a Geotab–McLeod integration on **16** curated **BigQuery** datasets, authoring dataset and chart-metric **SQL** to replace fragile raw-log regex parsing with typed data models
- Developed full stack AI Customer Health Dashboard using **React** and **Python**, aggregating data from **Pendo**, and **Salesforce** in **BigQuery** into account health score with LLM-backed CSM recommendations

SQL Server/Infrastructure DBA — Sep 2025 – Dec 2025
Sunlife — *Toronto, ON*

- Automated mapping of **400+ CIS** standards to **Imperva** scans with **OpenPyXL** in **Python** for **MongoDB, MSSQL, PostgreSQL, Oracle, MySQL**, reducing manual compliance review time by **70%**
- Developed **Splunk** dashboards for **MSSQL**, **Guardium Data Encryption**, **Oracle Key Vault**, creating **30+** panels to monitor security operational issues (e.g., suspicious logins, high traffic, backup errors)
- Scripted **Active Directory** functional ID expiry monitoring in **PowerShell**, querying attributes via **SQL** for **600+** accounts across **DEV, UAT, STAGE, PROD** environments

App Developer & Code Sensei — Jan 2025 – Apr 2025
Code Ninjas — *Richmond Hill, ON*

- Developed full-stack app using **Next.js**, **Tailwind CSS**, and **Firebase** to manage and display **50+** student session statistics—implemented **CRUD** functions on admin page, and grid display on session display page
- Standardized formatting of manually inputted times on **Excel** spreadsheet automatically using **regex** and **csv** in **Python**, to improve data entry time by **4x**
- Automated incomplete product information updates on **Shopify** using **Shopify API** for **400+** items
- Taught various levels of game development (scripting, tilemap and sprite design, etc.) on platforms such as **Microsoft MakeCode Arcade**, **Roblox Studio**, and **Unity**, to **70+** kids aged 5-12 every week

Projects

Communal Catalogue | *React, Firebase* — 2025

- Designed and developed a full-stack library management system using **React Vite** and **Firebase** empowering small communities to organize, track, and manage book checkouts easily
- Built a webcam ISBN scanner using **QuaggaJS**, allowing users to quickly checkout books by scanning barcodes, using **OpenLibrary API** to fetch book details
- Implemented role-based authentication system with **Firebase Auth** to manage admin access